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☰ Instructions

1.

Which of the following neural networks has a memory?

1 / 1 point
- ☒ LSTM

☐ 2D CNN

☐ 1D CNN

☐ ReLU

✔ Correct

Long Short-Term Memory (LSTM) networks have memory cells designed to retain information over time.
2.

A stack of perceptrons constitute a fully connected layer.

1 / 1 point
- ☐ False

☒ True

✔ Correct

A stack of perceptrons in a layer forms a fully connected layer, where each perceptron is connected to every neuron in the previous layer.
3.

A CNN is more explainable than an RNN.

1 / 1 point
- ☒ False

☐ True

✔ Correct

NN is a black box classifier.
4.

Which of the following steps can be taken to prevent overfitting in a neural network?

1 / 1 point
- ☐ Batch normalization

☐ Dropout of neurons

☐ Early stopping

☒ All of the above

✔ Correct

All these techniques are commonly used to prevent overfitting.
5.

A CNN is spatial, an RNN is temporal NN.

1 / 1 point
- ☐ False

☒ True

✔ Correct

CNNs handle spatial data, while RNNs handle temporal sequences.
6.

A Python list is order one Tensor.

1 / 1 point
- ☐ True

☒ False

✔ Correct

A vector is order one Tensor.
7.

Which of the following deep learning models uses back propagation?

1 / 1 point
- ☐ Multilayer Perceptron Network

☒ All of the above

☐ Recurrent Neural Network

☐ Convolutional Neural Network

✔ Correct

All of these models use backpropagation for learning.
8.

A matrix is order two Tensor.

1 / 1 point
- ☒ True

☐ False

✔ Correct

A matrix is a second-order tensor, representing a two-dimensional array of numbers.
9.

Conv3D can be used for video.

1 / 1 point
- ☒ True

☐ False

✔ Correct

Conv3D processes video by handling both spatial and temporal dimensions.
10.

A CNN is smaller size NN (size is number of model parameters) than a non-CNN network.

1 point
- ☒ False

☐ True

✘ Incorrect

CNNs often have fewer parameters than non-CNN networks due to parameter sharing in convolutional layers.